

Technical Document #104: Retrieval Augmented Generation - Comprehensive Overview

Technical Document #104: Retrieval Augmented Generation - Comprehensive Overview

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Key Takeaways:

- Query decomposition breaks complex queries into simpler sub-queries.
- Vector databases store dense embeddings for semantic search.
- Semantic search finds documents based on meaning rather than exact keywords.